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Algorithmic pricing: Effects on consumer trust and price search

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ABSTRACT

With the widespread use of pricing algorithms in online markets, prices are increasingly fluctuating, contradicting consumers' desire for price stability. This research examines the central form of algorithmic pricing in online markets—namely, algorithmic dynamic pricing (ADP). In five studies, including one real-world ADP encounter and four incentive-based experimental studies (in addition to two Web Appendix studies), we use price fairness and range–frequency theory to examine how ADP affects consumers' trust in ADP retailers and the extent of their price search. The findings reveal that ADP reduces trust in the ADP retailer, though this effect diminishes after consumers become accustomed to ADP. Furthermore, ADP prolongs price search duration, which lengthens with consumer ADP experience and shortens as ADP becomes the market norm. These findings suggest that retailers can implement ADP, as consumer backlash can be short-term. ADP retailers can also actively build consumers' trust and affect their search for prices through price-matching strategies. In particular, retailers can communicate their use of reactive ADP (i.e., ADP aligned with competitive prices) and offer price-matching guarantees.
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1. Introduction

Companies frequently rely on pricing algorithms to automate their price setting in online markets (Fisher et al., 2018); thus, algorithmic dynamic pricing (ADP) has become a common pricing practice in the digital era, with retailers regularly tweaking products' prices in their online shops (Eley, 2019). On Amazon.com, for example, millions of price changes occur on a daily basis, corresponding to a price change of approximately every 10 min for each product (Mehta et al., 2018). These price changes can be substantial: a price-monitoring provider found that on Valentine's Day 2015, the price of a Nikon camera changed within hours by up to 240 % on Amazon (Krämer & Kalka 2017).

Despite the prevalence of price changes, the question remains: how do consumers respond to ADP (Kopalle et al., 2023)? Recent studies indicate that consumers expect retailers to maintain stable prices over a reasonable period (i.e., static pricing) (Keller et al., 2022; Schlereth et al., 2018). Thus, ADP conflicts with consumers' preferences, disrupting their expectations of the retailer–consumer relationship. This misalignment could lead to distrust in ADP retailers, which poses a significant issue for them, as trust is a crucial foundation for consumers' purchasing decisions (Bleier & Eisenbeiss, 2015). Conversely, consumers might view ADP as an opportunity to find lower prices through increased price comparisons. This behavior could

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reduce retailers' margins as consumers identify and capitalize on lower prices (Gauri et al., 2008). Retailers need to comprehend how consumers perceive and react to ADP to design effective pricing strategies (Kopalle et al., 2023); if consumers lose trust and/or increase their price comparisons, the short-term benefits of ADP (e.g., stimulating sales and better skimming of willingness to pay) could be undermined. This leads to our first research question: *How does ADP affect consumers' trust in the ADP retailer and the extent of their price search?*

However, ADP is becoming increasingly widespread, prompting questions about consumer habituation to ADP. A few decades ago, only a few e-commerce giants pioneered ADP, testing a pricing tactic that "broke the norm" from a consumer perspective (Haws & Bearden, 2006; Kannan & Kopalle, 2001). Since then, studies have shown that prices in online markets have become more volatile and that many retailers are adopting ADP (Morris & Vincent-Humphreys, 2019). As a result, consumers may have become more habituated to ADP due to increased exposure and social acceptance (Spann et al., 2024). Initially strong negative reactions to ADP could potentially be mitigated by this habituation. Therefore, our second research question is: *How does consumers' habituation to ADP influence their trust in the retailer and the extent of their price search as ADP becomes more prevalent?*

In addition, many retailers often forgo introducing ADP and, consequently, its benefits out of anticipated consumer backlash (Spann et al., 2024). This raises the practically and theoretically relevant question of how retailers can actively mitigate consumer backlash (Vomberg, 2021). Price matching may significantly enhance an ADP retailer's value proposition from the consumer perspective by providing assurance of competitive pricing relative to other market players. The ADP context offers unique theoretical opportunities for exploring price matching. Traditionally, retailers have offered price-matching guarantees that allow consumers to obtain price adjustments when they find lower prices elsewhere (customer-initiated price matching) (Kukar-Kinney et al., 2007). In the ADP context though, retailers have an additional option: they can use a reactive ADP approach, monitoring competitor prices and adjusting their own prices accordingly (algorithm-initiated price matching) (Fisher et al., 2018). Leveraging this practice could further reassure consumers about competitive pricing. However, such price-matching strategies could also have drawbacks, such as overemphasizing the monetary aspects of transactions and potentially stimulating increased price searches (Srivastava & Lurie, 2001). These practices might ultimately backfire, undermining the intended benefits. Therefore, our third research question is: *How does ADP affect consumers' trust in the ADP retailer and the extent of their price search in the presence of price matching (reactive ADP or price-matching guarantees)?*

Although ADP is an important topic from the perspective of business practice, relatively few empirical studies explore consumer reactions to ADP (see Web Appendix A). Prior research provides valuable initial insights, revealing that consumers perceive ADP as unfair. However, a significant gap remains in understanding how ADP influences price search behavior. Furthermore, no studies to our knowledge have investigated whether consumers become accustomed to ADP and consequently react differently out of habituation. Finally, no studies have addressed the effects of price matching in the ADP context. In particular, previous ADP studies have not assessed whether price-matching guarantees are beneficial in the ADP context. Moreover, existing studies treat ADP algorithms as a black box, without investigating whether algorithm-based price-matching, in the form of reactive ADP, could mitigate adverse consumer reactions.

As such, the present research examines the implications of ADP on consumers' trust in the ADP retailer and price search across time. Consumers' trust in the ADP retailer reflects their belief that the retailer will act in their best interests (McKnight et al., 2002). Price search across time compares the prices of a product in the same store at different times, driven by a clear purchase intention rather than untargeted browsing. It involves waiting and re-visiting stores for a lower price; it can lead to accelerated purchases, purchase deferrals, and stockpiling (Gauri et al., 2008).

We conduct a series of studies involving nearly 3,250 people. First, we conduct a study in which consumers observe genuine ADP practices in the marketplace for products they actually purchase. Second, we rely on four incentive-aligned lab experiments (as well as two Web Appendix studies) and investigate ADP in a pre-purchase stage.

Overall, we contribute to the literature in three principal ways. First, we add to the underrepresented behavioral literature on price search (Koschate-Fischer & Wüllner, 2017). Our findings reveal that ADP stimulates price search, but this effect is context-dependent. Specifically, when ADP becomes the market norm, its impact on price search diminishes. By integrating range–frequency theory with the sparse literature on price search, we show how it accurately predicts when consumers compare prices and how ADP influences their search behavior. This addresses Spann et al.'s (2024) call to investigate consumer responses to price distributions.

Second, we add to the literature on the acceptance of pricing techniques by exploring habituation effects (Kahneman et al., 1986; Xia et al., 2004) and address the call for research on the influence of ADP on consumer behavior (Kopalle et al., 2023). We extend prior research on consumer reactions to ADP (Web Appendix A); we find that ADP can decrease consumer trust in retailers using ADP in the pre-purchase stage—the stage in the customer journey in which pricing information is paramount. Existing research focuses mostly on consumers' post-purchase reactions to ADP. To the best of our knowledge, our work is the first to demonstrate habituation to ADP—showing that consumers become more accepting of ADP after experiencing it or perceiving it as a market norm. These consumers likely gain a better understanding of the price-setting process and show a smaller decrease in trust when encountering ADP.

Third, we enrich theory by distinguishing between reactive ADP and price-matching guarantees and investigating their effects. From a managerial perspective, our findings show that ADP retailers can mitigate the adverse effects on consumer trust and price search through these price-matching strategies. Both reactive ADP and price-matching guarantees reduce the negative impacts of ADP. However, price-matching guarantees provide more consistent effects, whereas the effectiveness of reactive ADP is context-dependent. Our findings also indicate that consumer aversion is directed not at algorithmic pricing

itself but at the type of information the algorithm uses. Consumers react less negatively to reactive than proactive ADP. This insight is particularly relevant given the prevalence of reactive ADP among retailers (Fisher et al., 2018), highlighting its practical implications for improving consumer perceptions and acceptance of ADP.

2. ADP

Algorithmic pricing refers to data-analytics-based pricing algorithms that allow firms to automatically generate dynamic or personalized prices (Kopalle et al., 2023; Seele et al., 2021). We focus on ADP, or algorithmic pricing with respect to price differences over time (intertemporal prices) (Elmaghraby & Keskinocak, 2003). ADP is the most common form of algorithmic pricing in business practice (Walker, 2017) and, in light of recent privacy regulations (e.g., the European Union's General Data Protection Regulation), is likely to remain the dominant form in the future.

In essence, pricing algorithms are formulas used to determine prices. Compared with manual price setting, pricing algorithms lead to substantial changes in online retailers' prices; automated price setting enables retailers to constantly vary their prices, closely linking ADP to price fluctuations (Spann et al., 2024). The range (or amplitude) and frequency of price changes constitute ADP (Prakash & Spann, 2022). For example, US online retailers change the prices of 10 % of their products every day (with values ranging from 1 % to 37 %) (Brown & MacKay, 2023); even intraday changes are no exception (Alparicio et al., 2022; Brown & MacKay, 2023). Moreover, these changes can be substantial; their median absolute size is 11 % (Gorodnichenko et al., 2018). In addition, unlike other forms of price variations (e.g., discounts), changes from ADP are more frequent, are unpredictable, and do not follow traditional discount cycles. Prices based on algorithmic pricing systems are typically not predetermined; retailers typically do not announce such dynamic price changes (Spann et al., 2024).

However, consumers are becoming increasingly aware of ADP. Customers in today's marketplace have increasing insight into price development. They might have experienced a retailer's price adjustments or visited price comparison sites (Gunadi & Evangelidis, 2022), such as Keepa or CamelCamelCamel, which make historic prices available through, for instance, price charts (Web Appendix B1). For example, 4.5 million consumers visit Keepa per month on average (WebsiteIQ, 2023). Moreover, companies can proactively communicate the use of ADP (Prakash, 2023). Similarly, ADP represents a topic on which media outlets frequently report; as such, they expose ADP practices to broad audiences. Finally, consumers can learn about ADP from their reference groups and other consumers, as these groups are likely to share their experiences with and opinions of ADP. As ADP provokes focal changes in retailers' price setting (Brown & MacKay, 2023) and conflicts with consumers' preference for price stability (Keller et al., 2022), we explore how consumers respond to ADP.

3. Theoretical background

Our research focuses on the effects of ADP on consumer *trust* and *price search behavior*, which previous research identifies as the principal dependent variables in the ADP context (Kannan & Kopalle, 2001). Trust largely influences consumers' decision on where to buy, while price search mainly reflects their decision on how to buy. As consumers' trust evaluations depend on how fair they consider the retailer's price-setting approach, we first introduce price fairness theory. As price search likely depends on how attractive consumers find the offered price, we discuss range–frequency theory to understand consumers' evaluations of the offered price.

3.1. Price fairness theory as a foundation for consumer trust in a retailer

We draw on price fairness theory to understand why and how ADP likely affects consumers' trust in the retailer. Price fairness is an important foundation for consumer trust (Singh & Sirdeshmukh, 2000) and refers to a subjective sense of a price as right, just, or legitimate (Xia et al., 2004). Consumers judge the fairness of a price on the price paid and/or the procedure followed to set the price.

Evaluation of the price draws on distributive fairness; that is, consumers compare their outcomes (i.e., the offered price) with other outcomes (i.e., prices offered at competitors or at different points in time). Evaluation of the pricing procedure refers to procedural fairness, or whether the retailer "played fair" by complying with the rules or consumer expectations when setting the price. Procedural fairness implies that the price-setting process is consistent and represents the interests of all parties (Ferguson et al., 2014; Kukar-Kinney et al., 2007; Xia et al., 2004).

For the development of our hypotheses, an important point is that not all violations of distributive or procedural fairness provoke unfairness perceptions. According to the dual entitlement principle, consumers expect a reference price and grant companies a reference profit. If a firm's reference profit is threatened (e.g., by a cost increase), consumers may even consider higher prices fair (Kahneman et al., 1986). Similarly, consumers evaluate prices more fairly when prices result from factors beyond the retailer's control (e.g., market-wide shortage; Vaidyanathan & Aggarwal, 2003) or when the retailer's price setting does not take advantage of the customer (e.g., surge pricing; Bolton et al., 2003). Apart from knowing how consumers evaluate the fairness of a price, retailers need to understand how consumers evaluate the attractiveness of a price. How attractive a focal price is, compared with other expected prices, determines consumers' price search.

3.2. Range–frequency theory and price search

Consumers compare prices when the perceived benefits of price search are high (Gauri et al., 2008). Such benefits are likely high when consumers evaluate a currently offered price as less attractive than expected prices. Research suggests that price expectations are the main evaluation standard for consumers in markets with volatile prices and therefore are crucial in an ADP context (Jacobson & Obermiller, 1990). We draw on range–frequency theory (Parducci, 1965) to predict when consumers evaluate a current price as more attractive than expected prices. Range–frequency theory "is particularly relevant to environments in which price volatility is the norm" (Niedrich et al., 2009, p. 694).

Prior research has demonstrated that consumers' price evaluation resembles a contrast process (i.e., consumers compare a price with the range and rank of other prices) rather than an assimilation process (e.g., consumers compare a price with the average price) (e.g., Niedrich et al., 2001; 2009). More precisely, range–frequency theory indicates that a consumer's (i) price evaluation (E) is a weighted compromise (w) between a range (R) and a frequency (F) principle:

$$E_i = wR_i + (1 - w)F_i \tag{1}$$

Specifically, consumers judge the currently offered price (P_i) against the highest (P_{max}) and lowest (P_{min}) prices in their comparison set (i.e., distribution of expected prices)—the range principle (Equation (2)—and against the rank ($Rank_i$) of the offered price in relation to all prices in the comparison set (N)—the frequency principle (Equation (3):

$$R_i = (P_i - P_{min})/(P_{max} - P_{min}) \tag{2}$$

$$F_i = (Rank_i - 1)/(N - 1) \tag{3}$$

The resulting value from Equation (1) ranges from 0 to 1, where 0 implies that no other price in the comparison set is lower (i.e., most attractive) and 1 implies that the current price is the highest (i.e., least attractive) in the comparison set.

Table 1 illustrates how evaluations of the same price (e.g., $P_i = 100$) would alter depending on variations in the comparison set. According to the range principle, consumers evaluate the current price as less attractive if they expect a lower minimum ($R_1 > R_0$) or maximum price ($R_2 > R_0$) in the market and/or over time (Janiszewski & Lichtenstein, 1999). If the two prices move simultaneously but in different directions (i.e., consumers expect lower minimum and higher maximum prices), the net effect depends on which price effect consumers perceive to be stronger. Under the frequency principle, the same price becomes less attractive when consumers expect more instances of lower prices (i.e., higher rank: $F_3 > F_0$). The same rank is more attractive if consumers observe more price points ($F_4 < F_0$).

We draw on these considerations to develop our price search hypotheses. We predict how and when ADP affects consumers' anticipation of maximum and minimum prices, price ranks, and price points. We argue that more attractive prices will lower price search while more unattractive prices will increase price search.

4. Hypotheses development

Fig. 1 depicts our conceptual model. We explore consumers' reactions to ADP in terms of their trust in the retailer and price search. We examine price search across time, which is becoming increasingly important in ADP. Outside ADP, price search over time is generally less prevalent because prices are stable, and such search is only relevant when consumers expect planned promotions. However, ADP introduces random and uncommunicated price changes from the consumer's perspective, which makes price search across time particularly likely.

Moreover, we argue that habituation effects in the form of past ADP experience and ADP as the market norm moderate consumer responses. We also explore whether retailer price-matching strategies can affect consumer responses to ADP.

4.1. Effect of ADP versus static pricing on trust and price search

4.1.1. Trust

We expect ADP to lower consumers' trust in the retailer. Trust is a psychological state involving the acceptance of vulnerability. Consumers accept vulnerability if they expect the retailer to have positive intentions toward their welfare. Trustworthy retailers are considered caring, honest, and consistent in their dealings with customers, which may cause consumers to perceive transacting with them as less risky. Perceived price fairness is a foundational principle of consumer trust (Singh & Sirdeshmukh, 2000).

ADP likely reduces price fairness and thus decreases trust. Consumers' unfairness perceptions are strong when their reference transactions and implied norms are violated (Ferguson et al., 2014). ADP can conflict with consumers' expectations of how transactions should occur because it violates procedural fairness. Consumers typically expect prices to remain stable over a reasonable period (Haws & Bearden, 2006) and thus likely perceive ADP "as a breach of trust, and this may well harm the vendor's customer relationship" (Kannan & Kopalle, 2001, p. 71). Moreover, consumers are unlikely to infer that pricing algorithms dynamically vary prices to defend retailers' entitled profits, as this activity would imply, for example, intraday cost changes of consumer durable goods. Instead, consumers likely expect self-serving and profit-enhancing motives (Seele et al., 2021). We predict the following:

Table 1
Illustration of the range and frequency principles.

Scenario	P	P_{min}	P_{max}	Rank	N	R	F
Base	100	50	150	3	11	R_0 = 0.50	F_0 = 0.20
Variation 1 (range principle)	100	**25**	150	3	11	R_1 = 0.60	F_1 = 0.20
Variation 2 (range principle)	100	50	**125**	3	11	R_2 = 0.67	F_2 = 0.20
Variation 3 (frequency principle)	100	50	150	**7**	11	R_3 = 0.50	F_3 = 0.60
Variation 4 (frequency principle)	100	50	150	3	**21**	R_4 = 0.50	F_4 = 0.10

Notes: P = current price offered, P_{min} = minimum price, P_{max} = maximum price, Rank = order rank, N = number of different prices in the comparison set. R and F values result from Equations (2) and (3), respectively. Bold font denotes the variation relative to the base scenario (ceteris paribus).

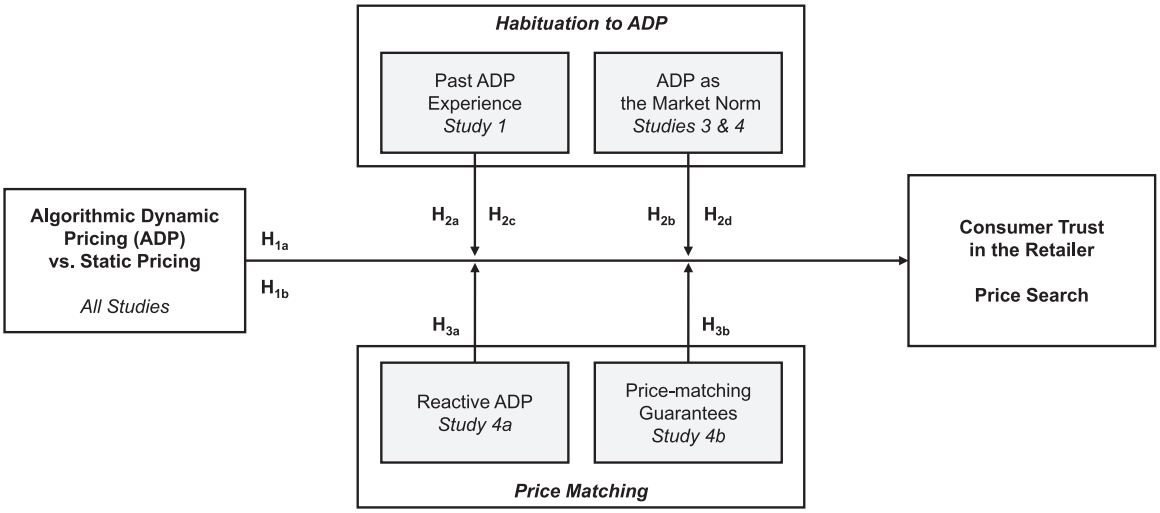

Fig. 1. Conceptual model.

H1a. ADP (vs. static pricing) reduces consumer trust in the ADP retailer.

4.1.2. Price search

ADP likely increases price search across time. ADP retailers tend to have more volatile prices whose future developments are uncertain for consumers (Drechsler & Natter, 2011). Prior studies show that more volatile prices increase consumers' expectations of finding lower prices (André et al., 2022; Kalwani & Yim, 1992). In addition, price uncertainty increases expected price ranges, particularly by increasing expected maximum prices (Mazumdar & Jun 1992, 1993).

According to the frequency principle, more instances of lower prices render the same price less attractive under ADP (vs. static pricing) (formally, the same price will have a higher rank under ADP; Table 1). Predictions from the range principle are inconclusive, as expected maximum prices are likely to increase while minimum prices are likely to decrease. However, given strong empirical evidence that price volatility lowers minimum price expectations (e.g., André et al., 2022; Drechsler & Natter, 2011), we expect the effect on minimum prices to be stronger. Thus, the range principle also implies that the same price becomes less attractive under ADP (vs. static pricing). We predict:

H1b. ADP (vs. static pricing) increases consumer price search across time.

4.2. Habituation effect: Past ADP experience and ADP as the market norm as moderators

Price fairness theory assumes that consumers habituate to different pricing practices: "'fairness' in a pricing arrangement depends heavily on consistency with the practices of the past to which people have become *habituated*" (Baumol & Fischer, 1987, p. 4, emphasis added). Research shows that such habituation can occur regardless of whether the original action was considered "legitimate" or "just" (Kahneman et al., 1986). Such habituation and adaptation can result from personal experience (Hoch & Deighton, 1989) or from following market norms (Melnyk et al., 2022). In the following, we discuss the moderating effects of past experience with ADP and ADP as the market norm.

4.2.1. Trust

First, past experience with ADP refers to a customer's prior exposure to and familiarity with ADP strategies. Such ADP experience can be direct, such as when customers notice that a retailer's prices vary frequently, and also indirect, such as

when media reports highlight the use and impact of ADP strategies. Customers with past experience are generally more familiar with the mechanics and rationale of ADP, resulting in less dissonance when confronted with ADP. They may have accepted ADP as an established "price rule" and changed their reference transactions. In such a case, fairness heuristic theory predicts that consumers will evaluate future transactions relative to this adjusted standard (Van den Bos et al., 1997). ADP will reduce procedural fairness for them to a lesser extent. ADP encounters will thus have less of an impact on trust in the retailer if consumers have already had experience with ADP.

H2a. The negative effect of ADP on consumer trust in the ADP retailer is weaker for consumers with past ADP experience.

Second, consumers' past experiences with ADP can influence not only their reference standard but also the extent to which they perceive ADP as the market norm. "ADP as the market norm" refers to general consumer acceptance of ADP as a competitive standard: many retailers employ ADP strategies in an industry, and customers recognize, understand, and generally accept the use of ADP as a common and expected aspect of their purchasing experience in the industry.

According to price fairness theory, a practice that becomes a norm is less likely to be perceived as unfair (Xia et al., 2004). Consumers consider the behavior of others a kind of "social proof" and adopt attitudes and actions under the premise that "[if] everyone is doing it, it must be a sensible thing to do" (Cialdini et al., 1990, p. 1015). If ADP is the market norm, consumers are likely to believe that an individual ADP retailer "plays by the rules" and is not in conflict with procedural fairness. Thus:

H2b. The negative effect of ADP on consumer trust in the ADP retailer is weaker when ADP is the market norm.

4.2.2. Price search

First, while past experience with ADP may mitigate negative effects on consumer trust in the ADP retailer, we hypothesize that it will prolong price search under ADP. Consumers with past ADP experience perceive less ambiguity when confronted with more volatile prices through ADP. They are more likely to perceive ADP not only as a risk but also as an opportunity to find lower prices due to higher price volatility (André et al., 2022). They also expect to find lower prices more frequently. According to range–frequency theory (Table 1), consumers who have already experienced ADP will perceive a focal price as less attractive and compare more prices when they encounter ADP. Thus:

H2c. The positive effect of ADP on price search is stronger for consumers with past ADP experience.

Second, when ADP is the market norm, more retailers are likely to change their prices more frequently. As a result, prices become more uncertain for consumers, which may lead them to expect higher prices (Mazumdar & Jun 1992, 1993). Consistent with this expectation, studies suggest higher prices due to possible collusion between competing firms through algorithmic pricing (Brown & MacKay, 2023; Calvano et al., 2020). Because consumers expect higher maximum prices, according to the range principle (Table 1), the prices of ADP retailers become more attractive if ADP is the market norm. They should therefore compare prices less. The frequency principle is less relevant at this point, as prices will change more in the market, though this does not necessarily affect the number of price changes of the focal ADP retailer (Brown & MacKay, 2023).

H2d. The positive effect of ADP on price search is weaker when ADP is (vs. is not) the market norm.

4.3. Can price matching mitigate adverse consumer reactions to ADP?

As the previous hypotheses imply, ADP leads to negative consumer reactions from the retailer's perspective. In the following, we derive hypotheses about whether retailers can actively mitigate negative consumer reactions with price matching.

Retailers could offer price-matching guarantees—that is, a commitment to match or beat the prices offered by competitors for the same product. This guarantee assures consumers that they will receive the most competitive price available in the market, either at the time of purchase or within a specified period after purchase. If a customer finds the same product offered at a lower price by a competitor, the retailer will adjust the purchased item's price to match the lower price or refund the price difference (Kukar-Kinney et al., 2007; Srivastava & Lurie, 2001).

An alternative ADP approach to ensure price matching is when retailers communicate that their algorithm automatically matches competitive prices. Many ADP retailers pursue reactive ADP—that is, price adjustments in response to external stimuli, particularly changes in competitors' prices (also referred to as repricing tools) (Spann et al., 2024). This approach leverages pricing algorithms that continuously and automatically match the prices of selected competitors up to a certain predefined lower price (Fisher et al., 2018). The hallmark of reactive ADP is that it positions the company as a price taker, passively responding to market changes initiated by competitors rather than proactively setting prices. By contrast, proactive algorithms dynamically set prices to exploit differences in price elasticities (at the market vs. individual consumer level). For example, they probe price elasticities at different times of the day or between weeks and set prices accordingly (Bradlow et al., 2017; Fisher et al., 2018). Alternatively, they take customer traffic as an input factor and adapt prices.

Price-matching guarantees and reactive ADP differ in their price-matching approach: price-matching guarantees are usually granted when customers request them. Furthermore, they are forward-looking assurances that allow customers to benefit from future price reductions by competitors within a certain period after purchase. Reactive ADP, by contrast, relies on

real-time data to automatically adjust prices without requiring direct customer participation; customers automatically benefit from competitive pricing. Unlike price-matching guarantees, reactive ADP focuses exclusively on the current situation and adjusts prices on the basis of current competitive prices without regard to future changes. However, both approaches also have similarities. Both strategies aim to maintain competitive prices compared with competitors. They offer consumers an assurance of competitive prices. We therefore derive the hypotheses jointly but test guarantees and reactive ADP in different studies.

4.3.1. Trust

We expect consumers to react more positively when ADP retailers offer price matching. Price matching can signal to consumers that the retailer has not developed its pricing algorithm to exploit customers opportunistically. Thus, price matching presumably offsets the procedural fairness violations caused by ADP. Previous research has shown that price-matching guarantees can promote fairness (Kukar-Kinney et al., 2007). We expect a similar effect in the ADP context.

Similarly, a reactive ADP approach can also build trust. Previous research has shown that consumers are more receptive to price increases that are due to factors external to the retailer and thus uncontrollable (Vaidyanathan & Aggarwal, 2003) and that do not take advantage of the customer (Campbell, 1999). Pricing algorithms based on competitors' prices will likely fall into this area.

H3a. The negative effect of ADP on trust in the ADP retailer is weaker if the retailer offers price matching.

4.3.2. Price search

We expect consumers to compare prices less when retailers offer price matching. Price matching gives consumers the sense that the retailer's prices are aligned with prices in the market. By offering to match prices, ADP retailers are signaling that their prices are below competitors' prices and that prices in the market are generally higher (Kukar-Kinney et al., 2007). This could give consumers the assurance they need to check prices less intensively. Knowing that they will not pay more than competitors' prices, consumers could simplify their decision-making process and reduce the time spent on comparing prices. In other words, price matching would lead to expectations of higher prices in the market and, as such, to fewer price comparisons. Thus:

H3b. The positive effect of ADP on price search is weaker when retailers offer price matching (vs. no price matching).

5. Overview of studies

We test our hypotheses in a series of five studies in which we measure and manipulate ADP encounters (Table 2). Study 1 is an observational study in which consumers find genuine ADP practices in the marketplace for products they actually purchase. Study 1 allows us to test habituation effects from past ADP experience. With this study, we investigate consumer reactions to ADP in "realistic shopping environments" (Haws & Bearden, 2006, p. 309)—an essential step for deriving managerial implications. In Study 2, we validate the findings of Study 1 in a more controlled setting. Study 3 investigates ADP as the market norm (vs. not market norm). In Studies 4a and 4b, price matching attenuates negative reactions to ADP. We explore two forms: communicating a reactive (vs. proactive) price-setting approach (Study 4a) and offering price-matching guarantees (Study 4b).

6. Study 1: Price-tracker study

Study 1 uses observational data to achieve two primary objectives: analyzing consumer reactions to real-life ADP and testing habituation effects based on past ADP experience. As learning from experience requires time to unfold, examining these effects within a realistic market context provides the most accurate insights.

6.1. Method

6.1.1. Participants and procedure

We recruited people who had purchased at least one product from a major global online retailer within the previous three months and invited them to participate in the study. Our screening criterion was not restrictive; all participants met this requirement. In a lab, we installed a price-tracking tool on all computers and asked participants to log in to their accounts with the online retailer to retrieve their most recent purchase. After doing so, they observed the price changes before and after their purchase date. Web Appendix B1 provides an example of the page participants saw, displaying a product with its price history over the past three months.

We administered a questionnaire to participants at three points in time: before viewing the price crawler results (t_0, pre-measurement of trust), immediately after viewing the results (t_1, online shopping experience), and four weeks after the lab session (t_2, post-measurement; trust and price search intentions). Introducing this time interval helps mitigate potential demand effects. Our sample consisted of 145 participants (51 % female; age range: 18–50 years).

Table 2
Overview of studies.

Study	Independent variables	Moderators	Dependent variables	Setting	Main findings: Trust	Main findings: Price search	Supported Hypotheses
S1	ADP (vs. static pricing)	Past ADP experience	Trust in retailer, price search intentions	Post-purchase, lab setting with actual purchases (N = 145)	• ADP reduces trust. Past ADP experience nullifies effects.	• ADP increases price search intentions. Past ADP experience amplifies the effect.	H_{1a}: ● H_{1b}: ● H_{2a}: ● H_{2c}: ●
S2	ADP (vs. static pricing)	–	Trust in retailer, price search across time	Pre-purchase, online experiment (N = 361)	• ADP reduces trust.	• ADP increases price search.	H_{1a}: ● H_{1b}: ●
S3	ADP (vs. static pricing)	ADP as the market norm (vs. not the market norm)	Trust in retailer, price search across time	Pre-purchase, online experiment (N = 424)	• ADP reduces trust. ADP as the market norm attenuates the effect.	• ADP increases price search. ADP as the market norm attenuates the effect.	H_{1a}: ● H_{1b}: ● H_{2b}: ● H_{2d}: ●
S4a	Reactive ADP (vs. proactive ADP vs. static pricing)	ADP as the market norm (vs. not the market norm)	Trust in retailer, price search across time	Pre-purchase, online experiment (N = 602)	• Reactive and proactive ADP reduce trust. ADP as the market norm attenuates the effect. Effect for reactive is weaker if ADP is the market norm.	• Reactive and proactive ADP stimulate price search. ADP as the market norm attenuates the effects. Effect is weaker for reactive ADP.	H_{1a}: ● H_{1b}: ● H_{2b}: ● H_{2d}: ● H_{3a}: ◐ H_{3b}: ●
S4b	ADP with price-matching guarantees (vs. ADP without guarantees vs. static pricing)	ADP as the market norm (vs. not the market norm)	Trust in retailer, price search across time	Pre-purchase, online experiment (N = 593)	• ADP with and without price-matching guarantees reduces trust. ADP as the market norm attenuates the effects.Effect is weaker for ADP with (vs. without) price-matching guarantees.	• ADP with and without price-matching guarantees increases price search. ADP as the market norm attenuates the effects.Effect is weaker for ADP with (vs. without) price-matching guarantees.	H_{1a}: ● H_{1b}: ● H_{2b}: ● H_{2d}: ● H_{3a}: ● H_{3b}: ●

Notes: ●: full support; ◐: partially supported (depending on condition); ○: no support.

6.1.2. Measures

We measured the dependent variables trust in the retailer with five items (α_{t0} = 0.87; α_{t2} = 0.90) and price search intentions with three items (α_{t2} = 0.89) (see Appendix B2 for detailed explanations). To assess the moderating variable, we asked participants if they had past ADP experience with the focal retailer (dummy coded).

We included several control variables. To capture different degrees of ADP, we included the range and frequency of price changes based on information from the price crawler. We asked participants to take screenshots to document their individual price crawler results. We calculated the range of price changes as the difference between the highest and lowest prices observed. Given that the price crawler scales the price history graphs to the price level, we did not focus on the relative price range but controlled for the price level. We also counted the number of price changes to measure the frequency of price changes. In addition, we accounted for the relative price position—that is, the degree of price advantage (or disadvantage) the consumer received over other prices from the price crawler. Finally, we controlled for the price paid for the purchased product, the product category, consumers' price consciousness, and their online shopping experience.

6.2. Results

6.2.1. Estimation and identification

We ensured the comparability of the treatment (ADP) and control (static pricing) groups, as demonstrated in Web Appendix B3, and present the regression equations in Web Appendix B2. To investigate the effects of ADP on trust, we use a

difference-in-differences (DID) model with observed control variables. The DID model addresses selectivity concerns that may arise from the non-random assignment of participants to the groups (i.e., static pricing or ADP) and controls for pre-study trust in the retailer.[1] As we measured some variables only in the ADP group (e.g., degree of price advantage), we do not include all lower-order interactive effects, which is standard practice in DID analyses. Given the lack of a pre-measurement for future price search intentions before exposure to the price crawler, we use ordinary least squares (OLS) regression rather than a DID model. Our control variables help alleviate potential selectivity concerns.

6.2.2. Trust

We estimate the DID models using OLS regressions with robust standard errors clustered at the participant level to account for within-subject correlation over time, which may arise from repeated measurement (Bertrand et al., 2004; Goldfarb et al., 2022). We standardized our independent variables. Models 1 and 2 in Table 3 display the results for consumer trust. In the DID analysis, the interaction between the group and post-period dummies ($\beta_{DID} = \beta_{ADP \times post}$) represents the treatment effect and thus is of focal interest.

We find that ADP reduces consumer trust in the retailer (model 1: $\beta_{DID} = -0.30$, $p < 0.05$), confirming H1a. Model 2 supports a habituation effect, in line with H2a. Past ADP experience attenuates the negative impact of ADP on consumer trust ($\beta_{DID \times Past_Exper} = -0.71$, $p < 0.05$). For consumers with past ADP experience, the effect of ADP becomes non-significant and is almost nullified ($\beta_{DID} + \beta_{DID \times Past_Exper} = -0.68 + 0.71 = 0.03$, $p = 0.89$).[2]

Regarding the covariates, participants in the ADP group observed different degrees of ADP (e.g., different numbers of price changes). We accounted for these differences by interacting the treatment effect with the range and frequency of price changes (model 2). We find that wider price ranges ($\beta_{DID \times Range} = -0.15$, $p < 0.05$) and more frequent price changes ($\beta_{DID \times Frequency} = -0.09$, $p < 0.05$) amplify the negative effects of ADP on trust, providing process evidence for our hypotheses (Goldfarb et al., 2022). In addition, while the degree of price disadvantage does not significantly affect trust ($\beta_{DID \times Disadvantage} = -0.02$, *ns*), the degree of price advantage ($\beta_{DID \times Advantage} = 0.28$, $p < 0.05$) significantly attenuates the negative effect of ADP on consumer trust.

6.2.3. Price search intentions

Models 3 and 4 in Table 3 display the results for price search intentions. In support of H1b, we find a positive relationship between ADP and consumer price search intentions (model 3: $\beta_{ADP} = 0.44$, $p < 0.05$). In line with H2c, consumers' past ADP experience enhances the impact of ADP on price search intentions (model 4: $\beta_{ADP \times Past_Exper} = 0.90$, $p < 0.05$). Specifically, consumers with past ADP experience report strong price search intentions ($\beta_{ADP} + \beta_{ADP \times Past_Exper} = 0.77$, $p < 0.05$), whereas the effect for consumers without past experience is non-significant ($\beta_{ADP} = -0.13$, *ns*).

Regarding the covariates, the results indicate that the range ($\beta_{ADP \times Range} = 0.22$, $p < 0.05$) and the frequency ($\beta_{ADP \times Freq} = 0.12$, $p < 0.05$) of price changes enhance the effect of ADP on price search intentions. This finding provides process evidence supporting our conceptual rationale (Goldfarb et al., 2022, p. 15). Moreover, neither the degree of being price-advantaged ($\beta_{ADP \times Advantage} = -0.10$, *ns*) nor that of being price-disadvantaged ($\beta_{ADP \times Disadvantage} = 0.00$, *ns*) affects the relationship between ADP and price search intentions.

6.3. Discussion

Study 1 demonstrates that genuine ADP encounters decrease consumer trust while enhancing price search intentions. It also reveals habituation effects: the negative impact of ADP on trust weakens for consumers with past ADP experience, whereas the effect of ADP on price search intentions becomes more pronounced. We ran a series of additional analyses and robustness checks, which we report in Web Appendix B4. We replicate our findings using an alternative measure of price search intentions. Finally, a single difference analysis (i.e., $trust_{t2} - trust_{t0}$ as the dependent variable) also corroborates our findings.

We did not observe significant effects based on the degree of price advantage or disadvantage. We attribute this to the study's post-purchase setting. People likely transfer the general pattern of ADP (range and frequency) to a new purchase situation, while the relative price position changes each time.

7. Study 2: Experimental replication in a pre-purchase setting

Although the findings from Study 1 support our hypotheses, the study has certain limitations. First, its design is correlational. Second, it focuses on the post-purchase stage, whereas findings might differ in the pre-purchase stage, in which pricing information is most pertinent. Third, Study 1 relies on self-reported measures of price search *intentions*. To address these limitations, Study 2 replicates the findings of Study 1 using a controlled experimental design in a pre-purchase setting, capturing actual price search *behavior*.

[1] We classified participants who observed at least one price change as the ADP group. Considering Haws and Bearden's (2006) findings, our classification is conservative; it works *against* finding effects of ADP. Including participants who are hardly treated might bias our estimates toward zero.

[2] The notation indicates a spotlight analysis of the two regression coefficients ($\beta_{DID} + \beta_{DID \times Past_Exper}$). The spotlight analysis documents the absolute effect for consumers with past ADP experience.

Table 3
Study 1: Effects of ADP on consumer trust and price search intentions.

		Consumer trust				Price search intentions			
		Model 1 $t_2 - t_0$		Model 2 $t_2 - t_0$		Model 3 t_2		Model 4 t_2	
		B	SE	*B*	*SE*	*B*	*SE*	*B*	*SE*
Overall effects									
ADP × post (DID)	H_{1a}	−0.30**	0.18	−0.68***	0.26				
ADP dummy (ADP)	H_{1b}	-0.00	0.20	0.23	0.31	0.44**	0.26	−0.13	0.35
Post-treatment dummy		−0.30**	0.17	0.08	0.20				
Habituation effects									
DID × Past ADP experience	H_{2a}			0.71**	0.35				
ADP × Past ADP experience	H_{2c}			−0.45	0.40			0.90**	0.47
Post-treatment dummy × Past ADP experience				−0.71**	0.29				
Characteristics of ADP									
MOD[a] × Degree of price advantage				0.28**	0.14			−0.10	0.14
MOD[a] × Degree of price disadvantage				−0.02	0.09			0.00	0.09
MOD[a] × Range				−0.15**	0.08			0.22**	0.11
MOD[a] × Frequency				−0.09**	0.04			0.12**	0.06
Control variables									
Past ADP experience		−0.05	0.18	0.35	0.34	−0.25	0.22	−0.92**	0.41
Price paid		0.14**	0.07	0.22***	0.06	0.06	0.10	−0.13	0.17
Product category dummy		0.22*	0.17	0.21*	0.17	−0.07	0.22	−0.05	0.21
Price consciousness		0.20**	0.10	0.19**	0.11	0.52***	0.11	0.51***	0.11
Online experience		0.20***	0.07	0.19***	0.07	−0.27***	0.09	−0.23***	0.09
Constant		5.22***	0.22	5.02***	0.27	4.71***	0.28	5.02***	0.31
R^2		0.14		0.19		0.17		0.22	

***$p < 0.01$; **$p < 0.05$; *$p < 0.10$ (one-tailed tests). [a]MOD = Moderator refers to DID in models 1 & 2 and to ADP in models 3 & 4.
Notes: Models 1 & 2 are DID models with robust standard errors clustered at the participant level. Models 3 & 4 are OLS regression models. Model 1 (3) shows the overall treatment effect of ADP (DID) on consumer trust (overall effect of ADP on price search intentions). Models 2 and 4 test habituation effects. We observe some variables (e.g., range of price changes) only for the ADP group.

7.1. Method

Participants were 361 people (50 % female; age range: 18–77 years) drawn from an online general population sample via Prolific, who completed an online survey for a nominal fee. We asked participants to imagine they were purchasing a TV and that they visited an online store where they found an LCD-TV with a 32-inch full-HD screen that met their expectations. All participants focused on the same TV throughout the study, eliminating selection variability.

We selected consumer electronics products for this and the subsequent studies because prior research has shown that price-setting procedures can be realistically manipulated in this context (Prakash & Spann, 2022). Prices for consumer electronics fluctuate over time, and there is little transparency in how companies set their prices (Ferguson et al., 2014). Participants had a budget of £400 for the purchase, with any remaining money available for future purchases. To ensure incentive compatibility, participants earned a bonus of £0.10 for every £10 left after purchasing the TV.

We first measured the dependent variable trust in the retailer with the same five items as in Study 1 ($\alpha_{t0} = 0.91$). We then randomly assigned participants to one of two conditions in an ADP versus static between-subjects experimental design. To manipulate the price-setting procedure, we told participants that from experience, they knew that prices in the online store are dynamic (static). They are not stable (are stable) over time, and, on top of some price adjustments, they frequently (hardly ever) change. Participants first observed an initial price for the TV (i.e., price on day 1) and were told that while they decided to buy this TV, they did not immediately make a purchase. Instead, they chose to visit the online store on subsequent days. They observed three more prices for the TV (learning phase: days 1–4),[3] taking as much time as they wanted to review the price each day before moving on to the next.

After observing the price on day 5, participants decided whether to purchase the TV at the offered price or to defer until the next day, hoping for a lower price. If participants delayed purchasing the TV until day 11, they had to buy it at the price offered on that day (purchase phase: days 5–11).[4] We used realistic price levels from actual stores and predetermined the

[3] We pretested this design and also tested a design featuring a learning phase of a single day. The manipulation was successful in both cases but was stronger in a four-day learning phase.

[4] This design limits the price search phase to seven days. Participants' knowledge that they can defer purchase at most six times might influence when they stop searching for a better price. In line with prior research (Jindal & Aribarg, 2021), we note that a participant's payoff from each price search decision depends solely on the trade-off between the cost of conducting one additional search and the incremental expected benefit of search. Prior knowledge of maximum searches allowed neither changes this trade-off nor affects the decision to price search.

sequence of prices (see Web Appendix C1). In the ADP (static) condition, prices changed frequently (did not change at all) during both the learning and the purchase phases (Web Appendix C2).

After participants purchased the TV, we repeated the trust measurement using the same items (α_{t1} = 0.95). We measured price search by the number of times participants deferred their purchase, hoping for a better price (price search across time). Moreover, participants responded to a manipulation check for ADP (vs. static pricing) (see Web Appendix C3 for all items) and to standard demographic questions.

7.2. Results

7.2.1. Manipulation check

The manipulation of the price-setting procedure was successful. Participants in the ADP condition believed less that the price would be the same the next day (M_{Static} = 6.22 vs. M_{ADP} = 2.04; t(359) = −43.20, $p < 0.01$).

7.2.2. Trust

As in Study 1, we estimated a DID model with robust standard errors clustered at the participant. In the DID analysis, the interaction between the group (ADP vs. static) and the post-period dummy ($b_{DID} = b_{ADP \times post}$) represents the treatment effect, our focal point of interest. The results show that ADP significantly reduces consumer trust in the retailer (b_{DID} = −1.71, $p < 0.001$), confirming H1a.

7.2.3. Price search

An OLS regression on price search, with the pricing approach (ADP vs. static) as the independent variable, showed that participants in the ADP condition engaged more in price search across time (M_{Static} = 4.37 vs. M_{ADP} = 5.03; b = 0.66, $p < 0.05$), confirming H1b. Note that we used OLS regressions to maintain consistency across Studies 1–4. In addition, a Poisson regression model replicates our findings (b = 0.14, $p < 0.01$). However, we focus on OLS regression in this and subsequent studies to enhance the interpretability of our effects.

7.3. Discussion

Using an incentive-compatible experimental design, we successfully replicate the main findings of Study 1 in a pre-purchase context. Our findings indicate that consumers react negatively to ADP, trusting ADP retailers less than static-pricing retailers. Furthermore, ADP stimulates increased price search behavior.

Building on the design of Study 2, we addressed additional questions in a Web Appendix study. In Web Appendix D, we investigated the role of a pricing algorithm setting prices automatically. We compared static pricing, manual dynamic pricing (a pricing manager sets prices), and automated dynamic pricing (an algorithm sets prices). Both manual and automated dynamic pricing reduce trust in the retailer and lead to increased price search compared with static pricing. However, our results do not indicate a general aversion to algorithms among participants, as the regression coefficients do not differ significantly between manual and automated dynamic pricing. The primary impact of the algorithm lies in creating significant price fluctuations that are unrealistic under manual dynamic pricing.

8. Study 3: ADP as the market norm

In Study 3, we explore competitive aspects, focusing on the impact of ADP when it becomes the market norm. As more retailers adopt ADP, consumers may become more accepting of it.

8.1. Method

Participants were 424 people (46 % female; age range: 18–79 years) drawn from an online general population sample via Prolific, who completed an online survey for a nominal fee. Consistent with Study 2, we selected the consumer electronics context for our study. We asked participants to imagine purchasing a laptop PC and visiting an online store to find a laptop that meets their expectations: a 14-inch screen, 16 GB RAM, and 256 GB SSD. Similar to Study 2, they had a budget (£680) for the purchase, and to ensure incentive compatibility, participants earned a bonus of £0.10 for every £10 they had left after buying the laptop.

We randomly assigned participants to one of four conditions in a 2 (price-setting procedure: ADP vs. static) × 2 (market norm: market norm vs. not market norm) between-subjects experimental design. In the market norm (no marketing norm) condition, we told participants that they are aware (not aware) of many (any) retailers whose prices are dynamic and frequently change. ADP is considered a new competitive standard, aligning with current market conditions (a departure from traditional pricing norms, raising consumer concerns about fairness). Consumers have accepted ADP as a fair and reasonable (distrust ADP as an unfair and manipulative) approach to pricing goods. They appreciate the responsiveness that ADP offers, as it allows for greater flexibility and better customization to their needs and preferences (they are frustrated when faced with fluctuating prices, as they feel they are being taken advantage of by retailers trying to maximize profits at their

expense). Retailers that adopt ADP strategies are often praised for offering competitive prices while maintaining profitability (face backlash, with consumers seeking predictable and transparent pricing elsewhere). Finally, we mentioned that, overall, ADP had become the standard (is not widespread) in the industry, and both retailers and consumers recognize its benefits and accept it as (drawbacks and consider it a deviation from) the market norm.

Our design mirrors Study 2 but includes a competitor: during the learning phase, participants observed prices at both the focal and a competing retailer from days 1 to 4. On day 5, participants could choose to check the competitor's price, buy the product at the focal retailer (either before or after seeing the competitor's price), buy the product at the competing retailer, or postpone their decision to the next day (either before or after seeing the competitor's price). If participants postpone the decision, they need to make the same decision the next day. They could postpone the decision until day 11. We used realistic price levels from actual stores and predetermined the price sequence (Web Appendix E1 & E2).

Before the manipulation, we measured the dependent variable trust in the retailer using the same five items as in Study 1 (α_{t0} = 0.91). After participants purchased the laptop, we repeated the trust measurement (α_{t1} = 0.95). We measured price search across time as in Study 2. Participants also responded to a manipulation check for ADP (vs. static pricing) (see Web Appendix E3 for all items) and a manipulation check for market norm (vs. not market norm). Finally, they provided standard demographic information.

8.2. Results

8.2.1. Manipulation checks

Two 2 × 2 analyses of variance (ANOVAs) with Bonferroni corrections for multiple testing of the manipulation check measures for ADP and market norm confirmed that the manipulations had the intended effects. For the ADP manipulation, only the main effect of ADP was significant ($F(1, 420) = 828.91$, $p < 0.01$). Neither the main effect of norm ($F(1, 420) = 1.20$, *ns*) nor the interaction ($F(1, 420) = 2.52$, *ns*) was significant. Participants in the ADP condition believed less that the price would be the same the next day ($M_{ADP} = 2.79$ vs. $M_{Static} = 6.30$). Similarly, for the market norm manipulation, only the main effect of the market norm was significant ($F(1, 420) = 220.98$, $p < 0.01$). The main effect of ADP ($F(1, 420) = 0.02$, *ns*) and the interaction effect ($F(1, 420) = 0.15$, *ns*) were not significant. Participants in the market norm condition agreed less that virtually no retailer uses ADP ($M_{Norm} = 2.37$; $M_{No\ Norm} = 4.78$).

8.2.2. Trust

We estimated a DID model with robust standard errors clustered at the participant. Model 1 in Table 4 replicates the effect observed in the previous studies and confirms H1a: the treatment effect of ADP is negative and significant ($b_{DID} = -1.56$, $p < 0.01$). In addition, model 2 shows that the interaction between the treatment effect and ADP as the market norm is significantly positive ($b_{DID \times Norm} = 1.04$, $p < 0.01$), confirming H2b. A spotlight analysis indicates that for consumers in the not market norm condition, ADP significantly reduces trust ($b_{DID} = -2.08$, $p < 0.01$). For consumers in the market norm condition, the effect of ADP is less negative but still significant ($b_{DID} + b_{DID \times Norm} = -1.00$, $p < 0.01$).

8.2.3. Price search across time

We conducted an OLS regression on price search across time with ADP and market norm as the independent variables and a set of dummy variables indicating whether consumers checked competitors' prices each day during the purchase phase (1 = consumer checked the competitor price on day X) to reduce bias in the estimation. Model 3 in Table 3 shows a significant, positive effect of ADP ($b_{ADP} = 1.48$, $p < 0.01$) on price search behavior, confirming H1b. A subsequent OLS regression (model 4) on the interactive effect of ADP and market norm on price search shows a significant effect ($b_{ADP \times Norm} = -1.82$, $p < 0.01$), in line with H2d. A spotlight analysis reveals that for consumers in the not market norm condition, ADP significantly increases price search across time ($b_{ADP} = 2.43$, $p < 0.01$). For consumers in the market norm condition, the effect of ADP is still significant but attenuated ($b_{ADP} + b_{ADP \times Norm} = 0.61$, $p < 0.01$).

8.3. Discussion

The findings of Study 3 show habituation to ADP. While consumers continue to trust ADP retailers less and compare prices more than they do with static pricing, the difference is less pronounced when ADP is the market norm.

In addition to price search over time, managers need to consider price search across different stores. This type of search is significant for retailers, as consumers comparing prices across stores are more likely to switch to competitors. A post hoc analysis (detailed in Web Appendix E4) reveals that ADP leads participants to compare prices between stores less frequently than under static pricing. One possible explanation is that participants expect the ADP retailer to offer lower prices than competitors. Moreover, as competitor prices fluctuate less, participants anticipate lower prices at those stores less often, which, according to range–frequency theory, leads to fewer price comparisons with competitors. This assumption is further supported by the finding that the negative effect of ADP on price search at other stores is attenuated when ADP is the market norm; if ADP is the norm, people are less likely to assume that the ADP retailer offers lower prices.

In Web Appendix F, we replicated Study 3 with an alternative manipulation of ADP as the market norm. Unlike Study 3, we did not refer to social reference groups but informed participants that many retailers (vs. virtually no other retailers) use ADP. The results indicate that the competitive standard alone is sufficient to reduce price search, similar to Study 3's findings.

Table 4
Study 3: Effects of ADP on consumer trust and price search behavior.

		Consumer trust				Price search behavior			
		Model 1 $t_1 - t_0$		Model 2 $t_1 - t_0$		Model 3 t_1		Model 4 t_1	
		B	SE	*B*	*SE*	*B*	*SE*	*B*	*SE*
Overall effects									
ADP × post (DID)	H_{1a}	−1.56***	0.11	−2.08***	0.15				
Market norm × post		0.19**	0.11	−0.33***	0.13				
ADP dummy (ADP)	H_{1b}	−0.03	0.08	−0.08	0.12	1.48***	0.15	2.43***	0.21
Market norm		0.07	0.08	0.03	0.11	−0.47***	0.16	0.48**	0.22
Post-treatment dummy		0.43***	0.08	0.66***	0.08				
Habituation effects									
DID × Market norm	H_{2b}			1.04***	0.21				
ADP × Market norm	H_{2d}			0.09	0.16			−1.82***	0.28
Control variables									
Competitor prices						Included		Included	
Constant		4.49***	0.07	4.51***	0.08	2.48***	0.16	1.97***	0.17
R^2		0.24		0.27		0.58		0.62	

***$p < 0.01$; **$p < 0.05$; *$p < 0.10$ (one-tailed tests).
Notes: Models 1 & 2 are DID models with robust standard errors clustered at the participant level. Models 3 & 4 are OLS regression models. Model 1 (3) shows the overall treatment effect of ADP (DID) on consumer trust (overall effect of ADP on price search). Models 2 and 4 test habituation effects.

However, there are no habituation effects for trust. This aligns with literature on norms (e.g., Opp, 1982), highlighting social reference groups' importance in norm formation and adherence.

9. Study 4a: Effects of automated price matching via reactive ADP

In Studies 4a and 4b, we test the effects of price matching. We use the two operationalizations we discussed previously: reactive (vs. proactive) ADP in Study 4a and price-matching guarantees in Study 4b.

9.1. Method

Participants were 602 people (48 % female; age range: 18–79 years) recruited from Prolific. As in the previous studies, we selected the consumer electronics context. We asked participants to imagine that they want to purchase a smartphone from an online store and that it meets their expectations (i.e., a 6.5-inch screen, 16 GB RAM, and 256 GB storage). Again, participants had a budget for the purchase (£380). To ensure incentive compatibility, participants earned a bonus of £0.10 for every £10 they had left after purchasing the smartphone.

We then randomly assigned them to one of six conditions in a 3 (price setting procedure: reactive ADP vs. proactive ADP vs. static) × 2 (market norm: market norm vs. not market norm) between-subjects experimental design. We manipulated ADP as the market norm exactly as in Study 3. We manipulated the price-setting procedure as in Studies 2 and 3 and extended it to differentiate between reactive and proactive ADP. As in the previous studies, we told participants that from experience, they know that prices in the online store are dynamic (static). They are not stable (are stable) over time, and, on top of some price adjustments, they frequently (hardly ever) change. In addition, we told participants in the ADP conditions that the focal retailer uses a reactive (proactive) ADP strategy. We described it as the retailer monitoring competitors' prices and copying them (conducting market experiments and setting its prices based on consumer demand). We also told participants that when the focal retailer detects that competitor prices are higher (consumers are willing to pay higher prices), it reacts by raising (proactively raising) its prices. By contrast, when it finds out that competitor prices are lower (consumers are more price-sensitive), the focal retailer reacts by lowering (proactively lowering) its prices. Last, we told participants that price-setting decisions are made automatically and immediately through the focal retailer's pricing algorithms, ensuring that prices always align with competitive factors (consumer preferences). We used realistic price levels from actual stores and pre-determined the price sequence (see Web Appendix G1 & G2). Participants followed the same procedure involving a learning phase and a purchase phase as in Study 3.

Before the manipulation, we first measured the dependent variable trust in the retailer using the same five items as in Study 1 ($\alpha_{t0} = 0.90$). After participants purchased the smartphone, we measured trust ($\alpha_{t1} = 0.95$) and price search across time as in Study 3. Participants also completed a manipulation check for ADP (vs. static pricing), a manipulation check for price matching (reactive ADP vs. proactive ADP vs. static pricing), and a manipulation check for market norm (vs. not market norm) (see Web Appendix G3 for all items). Finally, they responded to standard demographic questions.

9.2. Results

9.2.1. Manipulation checks

Three 3 × 2 ANOVAs with Bonferroni corrections confirmed that the manipulations had the intended effects. For the ADP manipulation, only the main effect of ADP was significant ($F(2, 596) = 443.87$, $p < 0.01$). Neither the main effect of norm ($F(1, 596) = 1.02$, *ns*) nor the interaction ($F(2, 596) = 1.02$, *ns*) was significant. Post hoc Tukey-HSD (honestly significant difference) tests indicated that participants in the two ADP conditions believed less that the price would be the same the next day ($M_{ADP_React} = 2.80$; $M_{ADP_Proact} = 2.78$; $M_{Static} = 6.20$) than participants in the static pricing condition (*p*s < 0.01). Similarly, in the price-matching manipulation, only the main effect of price-matching was significant ($F(2, 596) = 271.65$, $p < 0.01$). Neither the main effect of norm ($F(1, 596) = 0.08$, *ns*) nor the interaction ($F(2, 596) = 1.59$, *ns*) was significant. Participants in the proactive ADP condition agreed more that the retailer raises prices in response to consumer demand ($M_{ADP_React} = 4.74$; $M_{ADP_Proact} = 6.13$; $M_{Static} = 2.73$) than those in the other conditions (*p*s < 0.01). Finally, for the market norm manipulation, only the main effect of the market norm was significant ($F(1, 596) = 228.67$, $p < 0.01$). The main effect of ADP ($F(2, 596) = 2.14$, *ns*) and the interaction effect ($F(2, 596) = 0.20$, *ns*) were not significant. Participants in the market norm condition agreed less that virtually no retailer uses ADP ($M_{Norm} = 2.51$; $M_{No\ Norm} = 4.69$).

9.2.2. Trust

A DID model (model 1 in Table 5) reveals significantly negative treatment effects for both reactive ($b_{DID_Reactive} = -1.53$, $p < 0.01$) and proactive ($b_{DID_Proactive} = -1.64$, $p < 0.01$) ADP in line with H1a. Compared with static pricing, both forms of ADP decrease consumer trust in the ADP retailer. As hypothesized in H3a, the absolute value of the coefficient for reactive ADP is lower; however, the difference between the two coefficients is not statistically significant ($b_{DID_Reactive} - b_{DID_Proactive} = 0.11$; $p = 0.17$). Therefore, initial results provide no support for H3a.

Model 2 also considers the interactive effects of ADP as the market norm. As anticipated in H2b and consistent with Study 3, both interaction terms are significant, indicating habituation effects: ADP as the market norm mitigates the negative effects of both ADP forms ($b_{DID_Reactive \times Norm} = 0.79$, $p < 0.01$; $b_{DID_Proactive \times Norm} = 1.32$, $p < 0.01$). Despite this attenuation, both effects remain significantly negative ($b_{DID_Reactive} + b_{DID_Reactive \times Norm} = -1.13$, $p < 0.01$; $b_{DID_Proactive} + b_{DID_Proactive \times Norm} = -0.99$, $p < 0.01$).

However, model 2 (post hoc analysis) provides partial support for H3a. In line with H3a, the results indicate that when ADP is not the market norm, reactive ADP decreases trust less than proactive ADP ($b_{DID_Reactive} - b_{DID_Proactive} = 0.39$; $p < 0.05$). However, when ADP is the market norm, the difference is not significant ($[b_{DID_Reactive} + b_{DID_Reactive \times Norm}] - [b_{DID_Proactive} +- b_{DID_Proactive \times Norm}] = -0.14$; *ns*).

9.2.3. Price search across time

An OLS regression analysis (model 3) demonstrates that, consistent with H1b, both reactive ($b_{Reactive} = 1.22$, $p < 0.01$) and proactive ($b_{Proactive} = 1.51$, $p < 0.01$) ADP significantly stimulate price search behavior. However, reactive ADP stimulates price search to a lesser extent than proactive ADP ($b_{Reactive} - b_{Proactive} = -0.29$, $p < 0.05$). These results provide support for H3b.

In model 4, we further investigate the interactive effects of ADP as a market norm. As hypothesized in H2d and confirmed in Study 3, both interaction terms are significant, indicating habituation effects: ADP as a market norm reduces the positive effects of both ADP forms ($b_{Reactive \times Norm} = -1.33$, $p < 0.01$; $b_{Proactive \times Norm} = -1.21$, $p < 0.01$). Despite this attenuation, both effects remain significantly positive ($b_{Reactive} + b_{Reactive \times Norm} = 0.59$, $p < 0.01$; $b_{Proactive} + b_{Proactive \times Norm} = 0.95$, $p < 0.01$).

9.3. Discussion

In Study 4a, we examine the effects of automated price matching through reactive ADP. Consistent with the previous studies, we find that both reactive and proactive ADP decrease trust in the ADP retailer and lengthens the duration of price searches. The presence of ADP as the market norm moderates these effects, reducing their severity.

In addition, Study 4a sheds light on the role of the algorithm in pricing strategies. Significant differences emerged between reactive and proactive ADP. As anticipated, reactive ADP shortens the price search duration compared with proactive ADP. Thus, automated price matching in the form of reactive ADP can offer advantages for retailers. However, reactive ADP can only mitigate the loss of trust if ADP is the market norm. When ADP is not the market norm, participants appear less concerned with the specific algorithm the ADP retailer employs; the mere implementation of ADP diminishes their trust.

Our post hoc analysis in Web Appendix G4 reveals an initially counterintuitive finding: when ADP is not the market norm, reactive ADP prompts consumers to compare prices at competitors more than proactive ADP. This suggests that consumers associate reactive ADP with lower competitor prices in this context. However, after ADP becomes the market norm, reactive ADP reduces the extent of competitor price comparisons.

Table 5
Study 4a: Effects of reactive (vs. proactive) ADP on consumer trust and price search behavior.

		Consumer trust				Price search behavior			
		Model 1 $t_1 - t_0$		Model 2 $t_1 - t_0$		Model 3 t_1		Model 4 t_1	
		B	SE	*B*	*SE*	*B*	*SE*	*B*	*SE*
Overall effects									
Reactive ADP × post (DID_react)	H_{1a}	−1.53***	0.10	−1.92***	0.15				
Proactive ADP × post (DID_proact)	H_{1a}	−1.64***	0.11	−2.31***	0.14				
Market norm × post		0.54***	0.09	−0.16*	0.12				
Reactive ADP dummy (ADP_react)	H_{1b}	−0.02	0.08	−0.09	0.11	1.22***	0.16	1.93***	0.23
Proactive ADP dummy (ADP_proact)	H_{1b}	−0.09	0.08	−0.06	0.11	1.51***	0.16	2.16***	0.23
Market norm		0.04	0.07	0.01	0.12	−0.23*	0.14	0.68**	0.24
Post-treatment dummy		0.23***	0.07	0.58***	0.08				
Habituation effects									
DID_react × Market norm	H_{2b}			0.79***	0.20				
DID_proact × Market norm	H_{2b}			1.32***	0.20				
ADP_react × Market norm	H_{2d}			0.14	0.16			−1.33***	0.32
ADP_proact × Market norm	H_{2d}			−0.06	0.17			−1.21***	0.32
Control variables									
Competitor prices						Included		Included	
Constant		4.55***	0.07	4.55***	0.07	2.83***	0.16	2.34***	0.18
R^2		0.28		0.31				0.49	
						0.48			

***$p < 0.01$; **$p < 0.05$; *$p < 0.10$ (one-tailed tests).
Notes: Models 1 & 2 are DID models with robust standard errors clustered at the participant level. Models 3 & 4 are OLS regression models. Model 1 (3) shows the overall treatment effect of ADP (DID) on consumer trust (overall effect of ADP on price search). Models 2 and 4 test habituation effects.

10. Study 4b: Effects of price matching via price-matching guarantees

10.1. Method

Participants were 593 people (40 % female; age range: 18–82 years) recruited from Prolific. We gave participants the same purchase scenario, incentive alignment instructions, measurements of trust before ($\alpha_{t0} = 0.89$) and after ($\alpha_{t1} = 0.94$) being exposed to our manipulation, prices, and procedure involving a learning phase and a purchase phase as in Study 4a. In this study we randomly assigned participants to one of six conditions in a 3 (price matching: ADP with price matching vs. ADP without price matching vs. static) × 2 (market norm: market norm vs. not market norm) between-subjects experimental design.

We manipulated ADP as the market norm exactly as in Studies 3 and 4a. We manipulated the price-setting procedure as in Studies 2 and 3 and extended it to differentiate between offering a price-matching guarantee and offering no price-matching guarantee. We told participants in the ADP with price-matching guarantee condition that the focal retailer is confident it offers its customers the best prices on the market. However, in the event that a customer finds a better price within seven days after purchase, the retailer will match the price by returning the difference in cash. In the ADP without price-matching guarantee, we simply told participants that the focal retailer is confident it offers its customers the best prices on the market (Web Appendix H1 & H2).

Participants also completed a manipulation check for ADP (vs. static pricing), a manipulation check for the price setting procedure (ADP with price-matching guarantee vs. ADP without price-matching guarantee vs. static pricing), and a manipulation check for market norm (vs. not market norm) (see Web Appendix H3 for all items). Finally, they responded to standard demographic questions.

10.2. Results

10.2.1. Manipulation checks

Three 3 × 2 ANOVAs with Bonferroni corrections confirmed that the manipulations had the intended effects. For the ADP manipulation, only the main effect of ADP was significant ($F(2, 587) = 541.50$, $p < 0.01$). Neither the main effect of market norm ($F(1, 587) = 0.03$, *ns*) nor the interaction ($F(2, 587) = 2.93$, *ns*) was significant. Post hoc Tukey-HSD tests indicated that participants in the two ADP conditions believed less that the price would be the same the next day ($M_{ADP_Guarantee} = 2.87$; $M_{ADP_No\ Guarantee} = 2.78$; $M_{Static} = 6.50$) than participants in the static pricing condition (*ps* < 0.01). Similarly, in the price-matching manipulation, the main effect of price matching was significant ($F(2, 587) = 149.14$, $p < 0.01$, $\eta_p^2 = 0.34$). The main effect of market norm was not significant ($F(1, 587) = 0.10$, *ns*); the interaction was significant ($F(2, 587) = 4.94$, $p < 0.01$),

though its effect size was negligibly small ($\eta_p^2 = 0.02$)[5] (Perdue & Summers, 1986). Participants in the price-matching guarantee condition agreed more that the retailer refunds a price difference ($M_{ADP_Guarantee} = 5.58$; $M_{ADP_No\ Guarantee} = 3.12$; $M_{Static} = 3.02$) than those in the other conditions (*ps* < 0.01). Finally, for the market norm manipulation, the main effect of the market norm was significant ($F(1, 587) = 342.94$, $p < 0.01$, $\eta_p^2 = 0.37$). The main effect of ADP ($F(2, 587) = 7.30$, $p < 0.01$) was also significant; however, its effect size was negligibly small ($\eta_p^2 = 0.02$). The interaction effect ($F(2, 587) = 1.71$, *ns*) was not significant. Participants in the market norm condition agreed less that virtually no retailer uses ADP ($M_{Norm} = 2.26$; $M_{No\ Norm} = 4.73$).

10.2.2. Trust

A DID model (model 1 in Table 6) reveals significantly negative treatment effects for both ADP without ($b_{DID_No\ Guarantee} = -1.56$, $p < 0.01$) and with ($b_{DID_Guarantee} = -1.30$, $p < 0.01$) price-matching guarantees, consistent with H1a. Compared with static pricing, ADP reduces consumer trust in the ADP retailer. As hypothesized in H3a, the absolute value of the coefficient for ADP with price-matching guarantees is significantly greater ($b_{DID_Guarantee} - b_{DID_No\ Guarantee} = 0.26$; $p < 0.05$), confirming H3a.

Model 2 also considers the interactive effects of ADP as the market norm. As anticipated in H2b and consistent with Study 3, both interaction terms are significant, indicating habituation effects: ADP as the market norm mitigates the negative effects of ADP without ($b_{DID_No\ Guarantee \times Norm} = 0.72$, $p < 0.01$) and with ($b_{DID_Guarantee \times Norm} = 0.72$, $p < 0.01$) price-matching guarantees. Despite this attenuation, both effects remain significantly negative ($b_{DID_Guarantee} + b_{DID_Guarantee \times Norm} = -0.94$, $p < 0.01$; $b_{DID_No\ Guarantee} + b_{DID_No\ Guarantee \times Norm} = -1.20$, $p < 0.01$).

10.2.3. Price search across time

An OLS regression analysis (model 3) demonstrates that, consistent with H1b, both ADP with ($b_{Guarantee} = 0.84$, $p < 0.01$) and without ($b_{No_Guarantee} = 1.09$, $p < 0.01$) price-matching guarantees significantly stimulate price search behavior. However, ADP with price-matching guarantees stimulates price search to a lesser extent than ADP without guarantees ($b_{Guarantee} - b_{No_Guarantee} = -0.24$, $p < 0.10$). These results provide support for H3b.

In model 4, we further investigate the interactive effects of ADP as a market norm. As hypothesized in H2d and corroborated by Study 3, both interaction terms are significant, indicating habituation effects: ADP as a market norm reduces the positive effects of both ADP with ($b_{Guarantee \times Norm} = -1.24$, $p < 0.01$) and without ($b_{No_Guarantee \times Norm} = -1.20$, $p < 0.01$) price-matching guarantees on price search behavior. Only the effect of ADP without price-matching guarantees remains significantly positive ($b_{No_Guarantee} + b_{No_Guarantee \times Norm} = 0.50$, $p < 0.05$). The effect of ADP with price-matching guarantees no longer differs significantly from static pricing ($b_{Guarantee} + b_{Guarantee \times Norm} = 0.24$, $p < 0.27$).

10.3. Discussion

Study 4b explores whether consumer-initiated price matching through price-matching guarantees can mitigate the adverse effects of ADP. Consistent with previous studies, we find that ADP, whether it includes price-matching guarantees or not, diminishes trust in the retailer. However, ADP as the market norm attenuates these negative effects, demonstrating habituation effects.

Moreover, our findings highlight the beneficial impact of price-matching guarantees. As hypothesized, these guarantees significantly reduce the loss of trust in the ADP retailer and shorten the duration of price searches compared with ADP without such guarantees. Notably, when ADP is the market norm, price-matching guarantees result in no significant difference in price search duration length between ADP and static pricing.

By employing the same experimental design in Studies 4a and 4b, we facilitated direct comparisons between the effects of different price-matching strategies. Both reactive ADP and price-matching guarantees proved effective in mitigating the negative consequences of ADP. However, the effects of price-matching guarantees were even more pronounced; their influence was less dependent on the context, and their regression coefficients more closely resembled those of static pricing.

11. General discussion

This research investigates the impact of ADP on consumer trust and price search. Our findings indicate that ADP generally reduces trust and increases price search. However, habituation effects play a significant role: past ADP experience (Study 1) and ADP as a market norm (Studies 3to 4b) lessen the negative impact of ADP on trust. The effects on price search differ though: past ADP experience amplifies the impact of ADP on price search (Study 1), whereas ADP as a market norm reduces this effect (Study 3). In addition, we show that ADP retailers can mitigate the adverse effects on trust and price search through price-matching strategies. Both transparency in reactive ADP applications and price-matching guarantees are effective; they can diminish the impact of ADP on trust and price search (Studies 4a and 4b). Notably, our findings reveal that the effects of price-matching guarantees are less context-dependent.

[5] To demonstrate the orthogonality of the manipulation, only the main effect of price matching should be significant. Significant side effects can be disregarded if their effect sizes are substantially smaller than that of the primary effect of interest (Perdue & Summers, 1986). In our study, the effect size of the interaction is not only smaller but nearly zero as well ($\eta^2_{matching} = 0.34$ vs. $\eta^2_{interaction} = 0.02$); its significance is unlikely to be a threat to our results.

Table 6
Study 4b: Effects of ADP with price-matching guarantees on consumer trust and price search.

		Consumer trust				Price search behavior			
		Model 1 $t_1 - t_0$		Model 2 $t_1 - t_0$		Model 3 t_1		Model 4 t_1	
		B	SE	*B*	*SE*	*B*	*SE*	*B*	*SE*
Overall effects									
ADP w/ guarantee × post (DID_guar.)	H_{1a}	−1.30***	0.11	−1.66***	0.15				
ADP w/o guarantee × post (DID_noguar.)	H_{1a}	−1.56***	0.11	−1.92***	0.16				
Market norm × post		0.57***	0.09	0.09	0.13				
ADP w/ guarantee dummy (ADP_guar.)	H_{1b}	0.14	0.09	0.05	0.11	0.84***	0.16	1.47***	0.22
ADP w/o guarantee dummy (ADP_noguar.)	H_{1b}	0.11	0.08	0.19**	0.11	1.09***	0.16	1.70***	0.22
Market norm		−0.03	0.07	−0.04	0.12	−0.47***	0.14	0.38*	0.23
Post-treatment dummy		0.43***	0.08	0.67***	0.08				
Habituation effects									
DID_guar. × Market norm	H_{2b}			0.72***	0.21				
DID_noguar. × Market norm	H_{2b}			0.72***	0.21				
ADP_guar. × Market norm	H_{2d}			0.17	0.17			−1.24***	0.31
ADP_no_guar. × Market norm	H_{2d}			−0.15	0.16			−1.20***	0.31
Control variables									
Competitor prices						Included		Included	
Constant		4.40***	0.07	4.41***	0.08	3.22***	0.16	2.80***	0.18
R^2		0.19		0.21		0.46		0.48	

***$p < 0.01$; **$p < 0.05$; *$p < 0.10$ (one-tailed tests).
Notes: Models 1 & 2 are DID models with robust standard errors clustered at the participant level. Models 3 & 4 are OLS regression models. Model 1 (3) shows the overall treatment effect of ADP (DID) on consumer trust (overall effect of ADP on price search). Models 2 and 4 test habituation effects.

11.1. Theoretical contributions and future research opportunities

First, we extend research on price search, a crucial but underexplored construct in the behavioral pricing literature (Koschate-Fischer & Wüllner, 2017). We contribute to the understanding of how price volatility influences price search. While in general price volatility (range and frequency of price changes) drives price search (Urbany et al., 1996), our study reveals more nuanced findings. For example, we find no differences in the extent of price search between static pricing (i.e., no price volatility) and ADP (i.e., high price volatility) when ADP is the market norm. Price matching can also significantly mitigate the effects of price volatility.

We introduce range–frequency theory as a new theoretical perspective to explain how price volatility affects price search. Existing research primarily assumes that price variance reflects consumers' perception of price volatility; however, our findings suggest that range–frequency theory provides a more accurate understanding. On the one hand, research shows that consumers struggle more to learn the variance or standard deviation of a numerical distribution than to learn the mean (Beach & Scopp, 1968). Therefore, range–frequency theory may better reflect consumer perception. On the other hand, variance is a symmetric concept in which increases in expected minimum or maximum prices should comparably drive price search. By contrast, range–frequency theory suggests asymmetric effects: price search increases when expected minimum prices fall but decreases when maximum prices rise (see Table 1). Our observations align with this theory.

Our research also adds to the ongoing investigation of how ADP influences price expectations. Spann et al. (2024) highlight the need to understand which aspects of price distributions influence consumer perceptions of price magnitude. According to range–frequency theory, the end points (minimum and maximum prices) shape consumers' perceptions. We call for future research to further investigate how ADP shapes consumer price expectations and subsequent price search behavior. For example, future studies could evaluate expected price distributions (André et al., 2022). Replicating these effects in field studies and considering different product categories is also essential, given that consumers exhibit varying price sensitivities across categories (Homburg et al., 2019).

In addition, we focused on targeted price search across time. The literature on price search distinguishes this from incidental price search, which refers to unfocused browsing, and price search across different stores (Gauri et al., 2008). Future research could explore the effects of ADP on these other forms of price search. Our post hoc tests reveal intriguing initial effects regarding price search across stores. Specifically, we find that ADP leads to a decrease in the frequency of price comparisons among competitors. However, this effect is less pronounced with reactive ADP than with proactive ADP. Future research could investigate this phenomenon further, examining potential consumer switching behavior.

Second, we extend prior research on the acceptance of pricing mechanisms (Kahneman et al., 1986; Xia et al., 2004) by demonstrating habituation effects. Our studies show strong negative consumer reactions to ADP, such as decreased trust in the retailer (Haws & Bearden, 2006). Yet they also reveal habituation effects supporting the prediction that consumers "begin to expect dynamic prices, thereby mitigating any negative impact that might arise initially" (Kannan & Kopalle, 2001, p. 72).

Over time, consumers may develop an internal control mechanism, perceiving ADP as an opportunity to secure better prices, thus leading to greater acceptance.

Importantly, our study shows (see Web Appendix F) that habituation effects regarding consumer trust are absent when ADP is merely the competitive standard, adopted by many retailers. This finding aligns with research on norm development (e.g., Opp, 1982); for consumers to accept ADP, their social reference groups must deem it acceptable. Future research could investigate the establishment of such norms. For example, it would be valuable to examine whether acceptance of ADP on online marketplaces that are pioneers in ADP (e.g., Amazon) can pave the way for traditional online retailers to adopt and normalize ADP. While consumers may tolerate ADP on larger platforms (e.g., due to past experiences or the wide range of products that are otherwise difficult to manage), it is conceivable that they do not expect ADP from more traditional retailers. At least in the early stages, traditional retailers might face consumer resistance. Our assessment reflects recent developments: while ADP was previously discussed in the media primarily in relation to platforms like Amazon.com (e.g., Mims, 2017), attention is now increasingly on traditional retailers (e.g., Bennett, 2023). Future research should investigate how traditional retailers should navigate the initial phase of ADP adoption and how, or whether, habituation effects will occur (Vomberg, 2023).

Third, when studying ADP as the market norm, we explore how consumers react when pricing algorithms become ubiquitous. Prior research has shown that the widespread adoption of pricing algorithms can unintendedly result in algorithmic price collusion (e.g., Hansen et al., 2021). We document that the more widespread adoption of such algorithms stimulates consumers to compare prices less frequently. To our knowledge, research in the context of consumer responses to ADP (or personalized pricing) has not considered such competitor effects. Such effects are particularly important. On the one hand, most retailers at least partially base their ADP approach on competitor prices (Fisher et al., 2018). On the other hand, owing to price comparison portals, consumers can switch to a competitor when detecting lower prices. Therefore, we call for research to consider further competitive aspects. Studies could investigate consumer responses to ADP when competitors employ personalized pricing.

Fourth, by exploring the role of pricing algorithms, we extend prior research on human–algorithm interactions in marketplaces. Contrary to expectations (Spann et al., 2024), our findings indicate that it is not the use of algorithms per se that drives consumer reactions (in terms of trust and price search) to ADP but rather the associated price fluctuation. The study in Web Appendix D demonstrates that consumer reactions do not differ between experimental groups in which prices are set automatically by an algorithm versus manually by a pricing manager. Our finding is consistent with previous research that inextricably links ADP and price fluctuation (e.g., Brown & McKay, 2023). On a more general level, our findings do not replicate the general aversion to algorithms observed in other contexts (e.g., Dietvorst et al., 2018). However, such aversion tends to emerge when algorithms make errors, which was not the case in our study. In addition, as algorithms continue to evolve, consumers may become more receptive and accepting of their use, particularly as they gain more experience with and knowledge about these systems.

Our findings also highlight the significance of the information algorithms used. From a retailer's perspective, consumers respond more favorably to reactive ADP than to proactive ADP. Future research could investigate whether the automation of pricing by algorithms, particularly in the context of personalized pricing, alters consumer responses. It would also be valuable to examine whether reactions to personalized pricing vary depending on the type of information used as a basis for pricing decisions.

11.2. Managerial implications

Our study offers several valuable implications for retailers in the implementation and management of ADP (see Table 7). First, retailers can implement ADP without necessarily facing significant consumer backlash. Retailers are often hesitant to adopt ADP out of concerns about negative customer reactions (Spann et al., 2024; Vomberg, 2021). However, according to experts, static pricing is rarely optimal, and avoiding ADP can lead to lost margins (Bertini & Koenigsberg, 2024). Our findings suggest that negative customer reactions to ADP are typically short-term. As consumers gain experience with ADP and it becomes the market norm, habituation effects emerge, and negative reactions in terms of trust diminish.

Second, beyond avoiding consumer backlash, retailers must also focus on *building* trust and *managing* consumer behavior. Our findings indicate that ADP retailers should actively manage their communication to positively influence consumer perceptions (Table 7). The choice of the ADP approach is a strategic decision beyond our study's scope. However, if retailers opt for reactive ADP, they should communicate this effectively, emphasizing that they always offer competitive pricing through ADP. This approach contrasts with previous ADP communication strategies (Prakash, 2023), as it underscores a clear customer benefit (Bertini & Koenigsberg, 2024). In addition, retailers should train customer contact staff to justify ADP.

Third, retailers should offer price-matching guarantees regardless of the chosen ADP approach. These guarantees can significantly promote trust in the retailer and reduce price search length. A post hoc analysis revealed that the expected costs of offering price-matching guarantees are low. Retailers are unlikely to need to issue post-purchase price refunds frequently. In our study, price-matching guarantees did not increase post-purchase price search intentions ($M_{\text{No Guarantee}} = 4.13$ vs. $M_{\text{Guarantee}} = 4.17$; $t(391) = -0.21$, $p = 0.84$; item: "How likely is it that you would look at smartphone prices in the next seven days?" 1 = "very unlikely," 7 = "very likely"). In addition, while 63 % of participants stated that price-matching guarantees are common in online markets, 87 % have never used them.

Table 7
A manager-oriented comparison of the observed effects.

	Reactive ADP	Proactive ADP	PMG	No PMG	Fluctuation
Trust					
Habituation (low)	−−	−−	−−	−−−	−−
Habituation (high)	−	−	−	−−	0
Price search					
Habituation (low)	++	+++	++	+++	0
Habituation (high)	+	++	0	+	++

Notes: Effects are relative to static pricing: values from +++ (−−−) "significantly more (less) than with static pricing" to 0 "no difference to static pricing." PMG: price-matching guarantees. In the Fluctuation column, "Habituation" refers to past ADP experience (Study 1), while in the other columns, "Habituation" refers to ADP as the market norm. Fluctuation is the range and frequency of price changes. Web Appendix Fig. WB4.2 details the results.

Retailers can also grant price adjustments upon customer requests without issuing an official price-matching guarantee. This approach allows retailers to cater to different price-sensitive consumer segments based on self-selection. This strategy can be effective, especially when retailers match their own past prices (Kireyev et al., 2017).

Fourth, retailers should consider limiting the extent of price fluctuations (Table 7). Study 1 indicates that lower ranges and frequencies of price changes favorably influence customer responses. However, retailers must balance limiting ADP without significantly reducing expected profits.

CRediT authorship contribution statement

Arnd Vomberg: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Christian Homburg:** Writing – review & editing, Writing – original draft, Supervision. **Panagiotis Sarantopoulos:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijresmar.2024.10.006.

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